RILEY FOARD

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EDUCATION

Caltech, Mechanical Engineering (BS), Aerospace Engineering Minor *Sep 2025 – Present*

Robotics track, Caltech Baseball, awarded 2026 Caltech Athletics Rookie of the Year, awarded Jack Real Aero Club of Socal Scholarship. Shop and welding-certified at Caltech, Pomona, and Harvey Mudd. Significant experience in Python, MATLAB, SolidWorks, LabVIEW, Arduino, ROS, ANSYS, and COMSOL. 3.97 GPA.

Pomona College, Physics (BA), Mathematics Minor *Aug 2022 – May 2025*

Physics and Philosophy Club Founder, CSWIM Physics Thesis Writing Advisor, Physics TA, Innovation Lab Member. 3.86 GPA

WORK EXPERIENCE

Caltech, Perpall SURF Fellow, Turbulence and Combustion Research *Jan 2026 – Present*

- Designed and ran numerical simulations of Rayleigh-Taylor instabilities and turbulent combustion flows with Stampede3.
- Derived new theoretical models for confined turbulence that redefined how to simulate RT turbulence in different geometries.
- Results will be presented at the 2026 APS DFD Annual Meeting under the mentorship of Professor Guillaume Blanquart.

Caltech, Rover Mechatronics Team Project *Jun 2026*

- Built a $200 RC-controlled rover with rocker-bogie suspension, a 4-DOF robotic arm, and onboard computer vision in just 10 days.
- Led all electrical integration: Raspberry Pi + camera module, LiPo battery, RoboClaws, DC motors, servos, and RC communication.
- CAD-designed and built the robotic arm, implementing OpenCV computer vision for rock sample detection.
- Final robot successfully traversed rough terrain at 5 m/s, identified and retrieved the correct volcanic rock, and stored the sample.

Caltech, Space Engineering JPL Mission Design Project *Mar 2026 – Jun 2026*

- Structures & Thermal Systems Team Lead of a 30-student multi-asteroid rendezvous mission design course taught by former JPL Director Dr. Michael Watkins, serving as the only undergraduate on my team. 150 page report completed over 3 months.
- Built a spacecraft-wide ANSYS Thermal Desktop model to keep components within operational temperature limits.
- Managed spacecraft-wide mass budgets, FMEA, cross-team deliverables, and presented weekly on behalf of my team.

Caltech, Two-Stage Belt Drive Transmission Team Competition *May 2026*

- Achieved the fastest time to 250 RPM in 6 seconds and the 4th highest max wheel speed in the competition at 476 RPM.
- Performed all belt tensioning and gear ratio optimization for the design, drilled all holes and press-fitted all gears and bearings.

US Virgin Islands Office of Disaster Recovery, Engineering Internship *Jun 2025 – Aug 2025*

- Alongside USVI WAPA, “hurricane-proofed” the Donoe Solar Farm, preventing future multimillion dollar storm damage.
- Reinforced solar panel anchoring and foundations to withstand wind speeds up to 150 mph.
- Built a gimbaling prototype that senses wind patterns and orients panels to minimize uplift. To be implemented with future funding.

Harvey Mudd, Autonomous Underwater Vehicle Team Project *Mar 2025 – Jun 2025*

- Designed and deployed a $200 autonomous AUV at Dana Point to characterize harbor aquatic conditions, including pH, pressure, and light attenuation. Our team was recognized for the highest rated field dataset due to our attenuation factor of $0.59 \pm 0.03\ m^{-1}$.
- Led all electrical integration and waterproofing: Teensy microcontroller, LiPo battery, all sensors, DC motors, ESCs, circuitry.

Harvey Mudd, Formula SAE Founding Member *Oct 2024 – May 2025*

- Authored technical sections of the club proposal and secured 5,000 simulation core-hours through a sponsorship with SimScale.
- Developed a PVC pipe chassis prototype, and welded an aluminum chassis prototype alongside two other members.

Personal Project, Designing and Building a Mechanical Arm *Jun 2024 – Sep 2024*

- Designed, programmed, and built a 5-DOF robotic arm that mirrors joint motion using potentiometer data from a wearable frame.
- Iterated the design from 3D-printed prototypes to a machined aluminum system.

Pomona, Haptic Glove-Controlled Quadcopter Project *Jan 2024 – Aug 2024*

- Built a quadcopter controlled by wearable glove orientation using gyroscopic data, and altitude controlled by pressure sensors.
- Led all electrical integration: Arduino, power distribution, all sensors, custom radio communication, brushless motors, and ESCs.
- Co-programmed and calibrated the PID feedback loop, achieving stable hover and gesture-based flight control.

Pomona, SURP Fellow and Research Assistantship, COVID mRNA Stress Modeling *Oct 2023 – May 2025*

- Applied deep learning to optimize a phase mask for fluorophore PSFs to quantify mechanical stress in the pseudoknots.
- Assembled a hybrid TIRF-magnetic tweezer microscope, and designed and implemented the optical recombination pathway.
- Presented the work at Optica 2024 and the 2025 APS Global Physics Summit under mentorship of Professor Kaley McCluskey.